

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
ECE249	BASIC ELECTRICAL AND ELECTRONICS ENGINEERING	3	1	0	4	
Course Weightage Course Focus	ATT: 5 CA: 25 MTT: 20 ETT: 50 EMPLOYABILITY,SKILL DEVELOPMENT	Exam Category: 13: Mid Term Exam: All MCQ – End Term Exam: MCQ + Subjective				

Course Outcomes :Through this course students should be able to

CO1 :: Understand the workings of various semiconductor devices and analyze their applications.

CO2 :: Analyze virtual applications by configuring arduino board and sensor module.

CO3 :: Examine various number systems and their application in digital circuits.

CO4 :: Design combinational circuits with applications specific integrated circuits.

CO5 :: Develop sequential circuits with flip-flops and ICs for various applications.

CO6 :: Create skill-oriented projects with features and ergonomics.

	TextBooks (T)		
Sr No	Title	Author	Publisher Name
T-1	DIGITAL LOGIC & COMPUTER DESIGN	M. MORRIS MANO	PEARSON

	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	FUNDAMENTALS OF ELECTRICAL ENGINEERING AND ELECTRONICS	B.L.THERAJA	S. CHAND & COMPANY
R-2	. DIGITAL FUNDAMENTALS BY THOMAS L. FLOYD , R. P JAIN, PEARSON	THOMAS L. FLOYD , R. P JAIN	PEARSON
R-3	ELECTRONIC CIRCUIT FUNDAMENTALS AND APPLICATIONS	MIKE TOOLEY	NEWNES PUBLISHERS
R-4	DIGITAL ELECTRONICS PRINCIPLES,DEVICES AND APPLICATIONS	ANIL K. MAINI	WILEY

Other Reading (OR)	
Sr No	Journals articles as Compulsary reading (specific articles, complete reference)
OR-1	https://archive.nptel.ac.in/courses/117/106/117106086/ ,
OR-2	https://www.theengineeringprojects.com/2018/06/introduction-to-arduino-uno.html ,
OR-3	https://archive.nptel.ac.in/courses/108/108/108108122/ ,
OR-4	Introduction to semiconductor ,
OR-5	https://archive.nptel.ac.in/courses/108/105/108105132/ ,

Audio Visual Aids (AV)		
Sr No	(AV aids) (only if relevant to the course)	Salient Features
AV-1	https://www.youtube.com/watch?v=ul-xvww61Lo	Simulation of VI Characteristics of a Diode
AV-2	https://www.youtube.com/watch?v=65jaszUGxsw	Interfacing of LDR with Arduino on proteus software demonstration
AV-3	https://www.youtube.com/watch?v=fYzhpPNpK18	Interfacing of Ultrasonic sensor with Arduino on proteus software demonstration
AV-4	https://www.youtube.com/watch?v=Rn0llzRhg1g	K-map implementation on Proteus Software
AV-5	https://www.youtube.com/watch?v=PD-mPOa-jIY	Multiplexer implementation on Proteus Software
AV-6	https://www.youtube.com/watch?v=TkAe7pitTpI	Decoder implementation on Proteus Software
AV-7	https://www.youtube.com/watch?v=mf87dQ5LKIA	Implementation of flip-flops on Proteus Software
AV-8	https://www.youtube.com/watch?v=dmPOt76jjcI	Counter design on Proteus Software

Virtual Labs (VL)		
Sr No	(VL) (only if relevant to the course)	Salient Features
VL-1	http://vlabs.iitkgp.ac.in/be/exp5/index.html#	VI Characteristics of a Diode

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	7

Detailed Plan For Lectures

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Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples
Week 1	Lecture 1	Fundamentals of Electrical Laws, Semiconductor Devices and its Applications (Ohm's Law, Kirchhoff's Law)	R-1		Lecture1: Zero Lecture: Introduction of subject along with its practical relevancy Lecture2: Ohm's Law, Kirchhoff's Law	Students will understand the importance and applications of the course.	Power point presentation and Proteus Simulator	home wiring systems
	Lecture 2	Fundamentals of Electrical Laws, Semiconductor Devices and its Applications (Ohm's Law, Kirchhoff's Law)	R-1		Lecture1: Zero Lecture: Introduction of subject along with its practical relevancy Lecture2: Ohm's Law, Kirchhoff's Law	Students will understand the importance and applications of the course.	Power point presentation and Proteus Simulator	home wiring systems
	Lecture 3	Fundamentals of Electrical Laws, Semiconductor Devices and its Applications (Voltage division rule, Current division rule)	R-1		Voltage Division Rule, Current division rule	Student will learn about kirchhoff's laws, intuitive method of circuit analysis-series and parallel simplification,voltage and current division rule	Power point presentation, Lecturing and discussions	
Week 2	Lecture 4	Fundamentals of Electrical Laws, Semiconductor Devices and its Applications (Basics of semiconductors (Intrinsic and Extrinsic))		OR-3 OR-4	Basics of semiconductors (Intrinsic and Extrinsic)	student will learn about the various semiconductor devices	Lecture discussion followed by question and answer	
	Lecture 5	Fundamentals of Electrical Laws, Semiconductor Devices and its Applications (PN junction diode (working and characteristics) and its applications (rectifiers and switch))	R-1 R-3	AV-1 VL-1	PN junction diode (working and characteristics) and its applications (rectifiers and switch),	student will learn about the VI characteristics of PN diode	Proteus Simulator and Power point presentation	Power Supply
	Lecture 6	Fundamentals of Electrical Laws, Semiconductor Devices and its Applications (PN junction diode (working and characteristics) and its applications (rectifiers and switch))	R-1 R-3	AV-1 VL-1	PN junction diode (working and characteristics) and its applications (rectifiers and switch),	student will learn about the VI characteristics of PN diode	Proteus Simulator and Power point presentation	Power Supply

Week 3	Lecture 7	Fundamentals of Electrical Laws, Semiconductor Devices and its Applications (Bipolar junction transistor (types, modes, construction, and working CE configuration))	R-1 R-3		Bipolar junction transistor (types, modes, construction, and working CE configuration)	Students will understand about BJT and its applications	Proteus Simulator, Lecturing and discussions	speakers, Mobile Phones
	Lecture 8	Introduction of Arduino and Sensors(Analog and digital signals)	R-2		Analog and digital signals	Student learn about difference between analog and digital signal	Power point presentation, lecturing and discussions	Microphones, AM/FM Radio, Mobile Phones
	Lecture 9	Introduction of Arduino and Sensors(Arduino board (pin configuration and description))		OR-2	Arduino board (pin configuration and description)	Students will understand the Arduino board and its pin description	PowerPoint presentation and Proteus demonstration	
Week 4	Lecture 10	Introduction of Arduino and Sensors(Arduino board (pin configuration and description))		OR-2	Arduino board (pin configuration and description)	Students will understand the Arduino board and its pin description	PowerPoint presentation and Proteus demonstration	
	Lecture 11	Introduction of Arduino and Sensors(IR sensor.)		AV-2	IR sensor, LDR	Students will understand the applications of IR sensor and LDR.	Proteus Simulator	Automatic Street Lights, automatic doors, escalators
		Introduction of Arduino and Sensors(LDR)		AV-2	IR sensor, LDR	Students will understand the applications of IR sensor and LDR.	Proteus Simulator	Automatic Street Lights, automatic doors, escalators
	Lecture 12	Introduction of Arduino and Sensors(basic principle of ultrasonic sensor)		AV-3	basic principle of ultrasonic sensor, Temperature sensor (DHT11/DHT22)	Students will understand the applications of ultrasonic and temperature sensor	Proteus Simulator and Power point presentation	Parking Assistance, Line-Following Robots, Weather Monitoring
		Introduction of Arduino and Sensors(Temperature sensor (DHT11/DHT22))		AV-3	basic principle of ultrasonic sensor, Temperature sensor (DHT11/DHT22)	Students will understand the applications of ultrasonic and temperature sensor	Proteus Simulator and Power point presentation	Parking Assistance, Line-Following Robots, Weather Monitoring
Week 5	Lecture 13				Test			
	Lecture 14	Introduction to number system and logic gates (Number system (conversion))	T-1 R-2		Number system (conversion), codes (B-G, G-B, Excess-3, BCD)	Student will learn to convert from one number system to another system	Power point presentation and AI tool : Rapid table - base converter	Scientific calculator

Week 5	Lecture 14	Introduction to number system and logic gates(codes (B-G,G-B,Excess-3,BCD))	T-1 R-2		Number system (conversion), codes (B-G,G-B,Excess-3,BCD)	Student will learn to convert from one number system to another system	Power point presentation and AI tool : Rapid table - base converter	Scientific calculator
	Lecture 15	Introduction to number system and logic gates (Compliments, Binary Arithmetic (addition and subtraction using 2's complement))	T-1		Compliments, Binary Arithmetic (addition and subtraction using 2's complement)	Students will learn about complement method for binary arithmetic	Power point presentation, AI tool : calculatoronline.net	
Week 6	Lecture 16	Introduction to number system and logic gates(logic gates)	T-1 R-2 R-4		All basic logic gates AND, NAND OR, NOR XOR-XNOR and NOT and its truth tables	Students will understand these basic gates	Power point presentation and Proteus Simulator	
	Lecture 17	Introduction to number system and logic gates (boolean algebra)	T-1 R-2		Logic design using Boolean algebra	Student will learn about Boolean algebra and its function implementation	Power point presentation and Proteus Simulator	
	Lecture 18	Introduction to number system and logic gates(SOP and POS)	T-1 R-2	OR-1	SOP and POS Simplification	As a result of this students will learn Input combination for any cell can be easily determined and circuit simplification	Proteus Simulator, AI tool : boolean algebra solver	To reduce complexity of circuit
Week 7	Lecture 19	Introduction to number system and logic gates(K-Map (up to 4 variables))		AV-4	K- Map (up to 4 variables)	As a result of this students will learn Input combination for any cell can be easily determined and circuit simplification	Proteus Simulator and AI tool : kmap solver	To reduce complexity of circuit
		SPILL OVER						
Week 7	Lecture 20				Spill Over			
	Lecture 21				Spill Over			
		MID-TERM						
Week 8	Lecture 22	Introduction to Combinational Logic Circuits(Adders)	T-1 R-2		Adders (Half adder and Full adder)	As a result of this students will learn How to design and perform addition operation in ALU using Digital circuits	Proteus Simulator and Power Point Presentation	Calculators

Week 8	Lecture 23	Introduction to Combinational Logic Circuits(Subtractors)	T-1 R-2		Subtractors (Half subtractor and full subtractor)	As a result of this students will learn How to design and perform subtraction operation in ALU using Digital circuits	Proteus Simulator and Power Point Presentation	Calculators
	Lecture 24	Introduction to Combinational Logic Circuits(Comparators)	T-1 R-2		Comparator (1-bit) comparator)	As a result of this students will learn How to design and perform comparator operation in ALU using Digital circuits	Proteus Simulator and Power Point Presentation	calculator
Week 9	Lecture 25	Introduction to Combinational Logic Circuits(Multiplexers)	T-1 R-2	AV-5	Circuits used to handle different input and output variables for MUX, De-MUX design	As a result of this Digital circuits used to combine the data on a single channel and how to simplify the circuit designing	Proteus Simulator and Power Point Presentation	TDMA and FDMA in mobile communication
		Introduction to Combinational Logic Circuits(De-multiplexers)	T-1 R-2	AV-5	Circuits used to handle different input and output variables for MUX, De-MUX design	As a result of this Digital circuits used to combine the data on a single channel and how to simplify the circuit designing	Proteus Simulator and Power Point Presentation	TDMA and FDMA in mobile communication
	Lecture 26	Introduction to Combinational Logic Circuits(Multiplexers)	T-1 R-2	AV-5	Circuits used to handle different input and output variables for MUX, De-MUX design	As a result of this Digital circuits used to combine the data on a single channel and how to simplify the circuit designing	Proteus Simulator and Power Point Presentation	TDMA and FDMA in mobile communication
		Introduction to Combinational Logic Circuits(De-multiplexers)	T-1 R-2	AV-5	Circuits used to handle different input and output variables for MUX, De-MUX design	As a result of this Digital circuits used to combine the data on a single channel and how to simplify the circuit designing	Proteus Simulator and Power Point Presentation	TDMA and FDMA in mobile communication
	Lecture 27	Introduction to Combinational Logic Circuits(Decoders)		AV-6	Decoders and Encoder	Students will understand the applications of decoders and encoders.	Proteus Simulator and Power point presentation	BCD to Seven Segment displa
		Introduction to Combinational Logic Circuits(Encoders)		AV-6	Decoders and Encoder	Students will understand the applications of decoders and encoders.	Proteus Simulator and Power point presentation	BCD to Seven Segment displa

Week 10	Lecture 28	Introduction to Sequential Logic Circuits(Latch (SR and D))	T-1 R-2		Explanation of basic latch (SRLatch,D-latch)	As a result of this student will understand Basic working of SET RESET latch	white board and power point presentation	
	Lecture 29	Introduction to Sequential Logic Circuits(Flip-Flop (SR, JK, D and T))	T-1 R-2	AV-7	Explanation of all basic flip-flops circuits and its working (SR, JK, D and T) flip flops	Student will understand Basic working of SET RESET flip-flops,and its applications	Proteus Software and PowerPoint presentations	
	Lecture 30	Introduction to Sequential Logic Circuits(Flip-Flop (SR, JK, D and T))	T-1 R-2	AV-7	Explanation of all basic flip-flops circuits and its working (SR, JK, D and T) flip flops	Student will understand Basic working of SET RESET flip-flops,and its applications	Proteus Software and PowerPoint presentations	
Week 11	Lecture 31	Introduction to Sequential Logic Circuits(Master-Slave flip-flop)	T-1 R-2		Master Slave JK flip flop	Student will learn about the importance of Master Slave JK flip flop	Lecturing and discussion,	
	Lecture 32	Introduction to Sequential Logic Circuits(Conversion of basic flip-flop)		OR-5	Conversion of basic flip-flop	Students will learn the use of excitation table to do conversions	Brainstorming and lecturing	
	Lecture 33	Applications of Sequential Circuits(Registers: Operation of basic shift registers (SISO, SIPO, PISO, PIPO))	T-1 R-2 R-4		Introduction of all basic kinds of shift registers eg. SISO, SIPO, PISO, PIPO	Student will learn to design shift registers using combinations of D flip-flop	Power point presentation and Proteus simulator	computer keyboards and musical keyboards, Universal Serial Bus (USB)
Week 12	Lecture 34	Applications of Sequential Circuits(Registers: Operation of basic shift registers (SISO, SIPO, PISO, PIPO))	T-1 R-2 R-4		Introduction of all basic kinds of shift registers eg. SISO, SIPO, PISO, PIPO	Student will learn to design shift registers using combinations of D flip-flop	Power point presentation and Proteus simulator	computer keyboards and musical keyboards, Universal Serial Bus (USB)
	Lecture 35	Applications of Sequential Circuits(Counters: Asynchronous counter (UP/DOWN/Mod-N))	T-1 R-2		Asynchronous counter (UP counter/DOWN counter/Mod-N counter)	Student will learn to design counters and use them in real time applications	Power point presentation and Proteus Simulator	Wall clocks, Traffic light controllers
	Lecture 36	Applications of Sequential Circuits(Counters: Asynchronous counter (UP/DOWN/Mod-N))	T-1 R-2		Asynchronous counter (UP counter/DOWN counter/Mod-N counter)	Student will learn to design counters and use them in real time applications	Power point presentation and Proteus Simulator	Wall clocks, Traffic light controllers

Week 13	Lecture 37	Applications of Sequential Circuits(Synchronous counter (UP/DOWN/Mod-N))	T-1 R-2	AV-8	Synchronous counter (UP counter/DOWN counter/Mod-N counter)	Student will learn to design counters and use them in real time applications	Power point presentation and Proteus Simulator	Automated toll collection systems and Digital tachometers
	Lecture 38	Applications of Sequential Circuits(Synchronous counter (UP/DOWN/Mod-N))	T-1 R-2	AV-8	Synchronous counter (UP counter/DOWN counter/Mod-N counter)	Student will learn to design counters and use them in real time applications	Power point presentation and Proteus Simulator	Automated toll collection systems and Digital tachometers
	Lecture 39				Project			
Week 14	Lecture 40	Applications of Sequential Circuits(Ring counter and Johnson ring counter)	T-1 R-2		Design of Ring counter and Twisted Ring counterr	Student will learn how to combine the flip-flops to get shifting also through these counters	Power point presentation and Proteus simulator	LED chasers in decorative lighting,
		SPILL OVER						
Week 14	Lecture 41				Spill Over			
	Lecture 42				Spill Over			
Week 15	Lecture 43				Spill Over			
	Lecture 44				Spill Over			
	Lecture 45				Spill Over			

Scheme for CA:

CA Category of this Course Code is:A0203 (2 best out of 3)

Component	Weightage (%)	Mapped CO(s)
Project	50	CO1, CO2, CO3, CO4, CO5, CO6
Test	50	CO1, CO2

Details of Academic Task(s)

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Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Project	To assess students' ability to design, implement, and troubleshoot electronic circuits, demonstrating their practical skills and problem-solving capabilities.	Expectations- (Topic selection, Learning outcome, teamwork): 25 Marks Working Model: 20 Marks Project Exhibition: 20 Marks Viva Voce: 15 Marks Ergonomics: 10 Marks Home assignment questions: 10 MARKs	Group	Offline	100	3 / 13
Test	To assess the ability to understanding and practical application of electrical laws, semiconductor devices, and sensor-based systemsor	Total: 30 MCQ-based questions, each of 1 mark with 25% negative marking	Individual	Offline	30	4 / 5

Plan for Tutorial: (Please do not use these time slots for syllabus coverage)

Tutorial No.	Lecture Topic	Type of pedagogical tool(s) planned (case analysis,problem solving test,role play,business game etc)
Tutorial1	Ohm’s Law, Kirchhoff’s Law, Voltage and Current division rule	Problem Solving
Tutorial2	Activity1	Proteus Installation and demonstration
Tutorial3	PN junction diode and bipolar junction transistor	Problem Solving
Tutorial4	Activity2	Flipped Classroom Activities
Tutorial5	Integration with sensors, Number system, Boolean algebra	Problem Solving
Tutorial6	SOP,POS and K-Map	Problem Solving
Tutorial7	Activity3	Pictionary-based Activities
After Mid-Term		
Tutorial8	Adder and Subtractor	Problem Solving
Tutorial9	Multiplexer and decoder	Problem Solving
Tutorial10	Activity4	Flipped Classroom Activities
Tutorial11	Flip-flop conversion and Shifting operation in digital design	Problem Solving

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Tutorial12	Counter design (Asynchronous Counter and Synchronous Counter)	Problem Solving
Tutorial13	Activity5	Test,Project exhibition
Tutorial14	Activity6	Test,Project exhibition

